1.2 Inverse of a Non-Singular Square Matrix

We recall that a square matrix is called a **non-singular matrix** if its determinant is not equal to zero and a square matrix is called **singular** if its determinant is zero. We have already learnt about multiplication of a matrix by a scalar, addition of two matrices, and multiplication of two matrices. But a rule could not be formulated to perform division of a matrix by another matrix since a matrix is just an arrangement of numbers and has no numerical value. When we say that, a matrix A is of order n, we mean that A is a square matrix having n rows and n columns.

In the case of a real number $x \neq 0$, there exists a real number $y\left(=\frac{1}{x}\right)$, called the inverse (or reciprocal) of x such that $xy = yx = 1$. In the same line of thinking, when a matrix A is given, we search for a matrix B such that the products AB and BA can be found and $AB = BA = I$, where I is a unit matrix.

In this section, we define the inverse of a non-singular square matrix and prove that a non-singular square matrix has a unique inverse. We will also study some of the properties of inverse matrix. For all these activities, we need a matrix called the adjoint of a square matrix.

1.2.1 Adjoint of a Square Matrix

We recall the properties of the cofactors of the elements of a square matrix. Let A be a square matrix of by order n whose determinant is denoted $|A|$ or $\det(A)$. Let a_{ij} be the element sitting at the intersection of the ith row and j^{th} column of A. Deleting the ith row and j^{th} column of A, we obtain a sub-matrix of order $(n-1)$. The determinant of this sub-matrix is called **minor** of the element a_{ij}. It is denoted by M_{ij}. The product of M_{ij} and $(-1)^{i+j}$ is called **cofactor** of the element a_{ij}. It is denoted by A_{ij}. Thus **the cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$.**

An important property connecting the elements of a square matrix and their cofactors is that the sum of the products of the entries (elements) of a row and the corresponding cofactors of the elements of the same row is equal to the determinant of the matrix; and the sum of the products of the entries (elements) of a row and the corresponding cofactors of the elements of any other row is equal to 0. That is,

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} = \begin{cases} |A| & \text{if } i = j \\ 0 & \text{if } i \neq j, \end{cases}$$

where $|A|$ denotes the determinant of the square matrix A. Here $|A|$ is read as "determinant of A" and not as " modulus of A". Note that $|A|$ is just a real number and it can also be negative. For instance, we have $\begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{vmatrix} = 2(1-2) - 1(1-2) + 1(2-2) = -2 + 1 + 0 = -1.$

Definition 1.1

Let A be a square matrix of order n. Then the **matrix of cofactors** of A is defined as the matrix obtained by replacing each element a_{ij} of A with the corresponding cofactor A_{ij}. The **adjoint matrix** of A is defined as the transpose of the matrix of cofactors of A. It is denoted by adj A.

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